Principal component analysis (PCA)

Launcher: `Toolbox/pca.bat`

Purpose

PCA transforms correlated spectral bands into uncorrelated principal components. Early components usually capture broad brightness and vegetation variation; later components may expose subtle crop or soil contrasts, but may also be dominated by noise. This tool repeats PCA for several band combinations.

Inputs and parameters

Option	What to provide	Behaviour
Input image	An 8-band WV3 or LEGION GeoTIFF	Band positions are interpreted using the selected sensor. A corrected or pansharpened reflectance product is normally appropriate. Images with NoData (for example, clipped in an earlier step) are supported: see NoData in the input image.
Results folder	Existing folder, or blank	Blank creates PCA in the input image's grandparent directory. A manually selected folder must already exist.
PCA combinations CSV	Optional CSV	Blank uses the curated combinations for the selected sensor (10 combinations, see below). Supplying a CSV processes only its listed combinations.

Option	What to provide	Behaviour
Run all band combinations	Checkbox, off by default	Processes every 3-, 4-, 5-, 6-, and 7-band combination plus the full 8-band set (219 runs). The output is very large. Cannot be combined with a CSV.
Vector layer	Optional clip polygon	Crops each PCA product after calculation and assigns NoData `−9999` outside the polygon.
Sensor	`WV3` or `LEGION`	Defines the eight band names and their positions, and which curated combinations are used. Default: `WV3`. See the band table.
Output compression	`Lossless` or `Near-lossless`	Lossless (default) is bit-exact. Near-lossless (LERC) is about 2.5× smaller, with an error of at most 1e-3 per pixel. Use a lossless input image: PCA can amplify errors carried from near-lossless inputs. See Output compression.

Curated combinations

id	WV3	LEGION
23	B-G-R	B-G-R
42 / 43	G-R-N	G-R-N
53 / 54	R-RE1-N	R-RE1-N
56	RE1-N-N2	RE1-RE2-N
97 / 98	B-G-R-N	B-G-R-N
122 / 123	Y-R-RE1-N	Y-R-RE1-N

id	WV3	LEGION
126	R-RE1-N-N2	R-RE1-RE2-N
127	C-B-G-Y-R	C-B-G-Y-R
134 / 135	C-B-G-RE1-N	C-B-G-RE1-N
210	G-Y-R-RE1-N-N2	G-Y-R-RE1-RE2-N
219	all 8 bands	all 8 bands

> **How the set was chosen.** Cropmarks mostly appear as differences in plant vigour or soil colour, so the set covers the three spectral regions where those differences show:
>
> - **Visible** (soil and plant pigment colour): B-G-R, the visible-only C-B-G-Y-R.
> - **Red edge** (the region most sensitive to chlorophyll and crop stress): R-RE1-N, Y-R-RE1-N, R-RE1-N-N2, and RE1-N-N2 (LEGION: RE1-RE2-N).
> - **Near infrared** (canopy structure and water stress): G-R-N, R-RE1-N-N2.
>
> Other criteria:
>
> - G-R-N and B-G-R are the standard colour-infrared and true-colour composites. Their components can be read against those familiar views.
> - B-G-R-N is the four-band VNIR set common to most satellites, for comparison with other imagery.
> - G-Y-R-RE1-N-N2 drops the coastal and blue bands, which carry the most haze left after atmospheric correction. The full 8-band set keeps every contrast PCA can find.
> - LEGION has a second red-edge band (RE2) where WV3 has a second NIR band (N2), so the LEGION file uses RE2 in its place.
> - Ten combinations keep a default run at 30 output bands.
>
> All the previous combinations were a starting point. The combination 134 (LEGION: 135) was also validated on cropmark data, discretizating the cropmarks really well.

The files are `Toolbox/app/sensors/data/wv3_pca_combinations.csv` and `legion6_pca_combinations.csv`; copy one as a starting point for your own CSV. Their ids are the same as in the full sweep, so `COMB<id>.tif` holds the same bands whichever mode produced it. In your own CSV, use ids that are new to the output folder (see Outputs).

To look up the id of any combination, see `wv3_pca_all_combinations.csv` and `legion6_pca_all_combinations.csv` in the same folder. They list the full sweep (all 219 combinations) in the same format. You can also copy rows from them into your own CSV.

Combinations CSV format

The file must have these exact columns:

```
id,bands,positions
1,C-B-G,1-2-3
2,R-RE1-N,5-6-7
```

`id` becomes the output number; `bands` is descriptive; `positions` controls the actual one-based source bands. Separate values with hyphens. An invalid file causes processing to stop. The full sweep of an eight-band image contains 219 combinations, so it creates 219 potentially large raster products.

Processing behaviour

For every combination, the tool selects the requested bands, standardizes them (centres and scales them), and runs Orfeo ToolBox PCA. It keeps the first three components (or fewer, if a combination has fewer bands): they hold most of the variance and fill an RGB view. Processing uses 2048 MB as the OTB streaming RAM setting.

NoData in the input image

A pixel that is NoData in any band of a combination is left out of that combination's PCA and is NoData (`-9999`) in its output.

Orfeo ToolBox does not read the NoData value of an image: it would use it (for example `-9999`) as if it were a measurement. The first component would then only separate the NoData area from the data, and one of the three components would be lost. To avoid this, the tool replaces the NoData pixels with the mean of each band before running the PCA, as the pansharpening tool does for its Bayesian method. A pixel at the band means sits at the centre of the data, so it changes neither the band means nor the directions of the principal components: the components are the same as those of a PCA computed from the valid pixels only.

The only effect is on the scale of the values. The components of an image without NoData have a standard deviation of 1. With NoData, the standard deviation over the valid pixels is `1 / sqrt(valid fraction)`: about 1.41 when half of the image is NoData. This factor is the same for every component, so it does not change how a component looks; keep it in mind only when comparing values between images.

Outputs

- `band_combinations.csv`: every combination run in this folder, across all runs
- `COMB<id>.tif` for each combination, with three bands (PC1–PC3)

Outputs are floating-point Cloud Optimized GeoTIFFs with NoData `−9999`: DEFLATE with the floating-point predictor (lossless), or LERC (near-lossless). Temporary files get random names and are deleted at the end of each combination, even if the run fails. Existing `COMB<id>.tif` products are skipped, which allows a long run to resume. Before anything runs, the tool checks each existing `COMB<id>.tif` against `band_combinations.csv`: if the CSV says it holds different bands than the requested combination `<id>`, or has no entry for it, the run stops and lists the clashing ids. Use another output folder, or give your combinations unused ids. Other settings (compression, clip vector, input image) are not checked.

Interpretation

Use `band_combinations.csv` to identify the source bands behind each result. Inspect every component, not just component 1. A component's sign is arbitrary: the same contrast can appear light in one run and dark in another. PCA magnitude is not an archaeological probability, and components from different combinations are not directly comparable without examining their loadings and histograms. Validate anomalies against original bands, spectral indices, topography, field patterns, and other dates.